Grug (30000 BCE) finally know why long bow gooder than thick bow

ME 80 Final Project

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Abstract

Grug know long bow gooder than thick bow. Grug want know why. Grug like math. Grug (somehow) know CAD. And, most important to Grug, Grug has bow to experiment with. Grug happy. Grug history's first mechanical engineer.

1 Introduction

For thousands of years, humans of different societies and civilizations have independently discovered the bow and arrow, like our curious friend Grug. Through this project, we aim to use modern structural mechanics and simulation techniques to optimize the design of a standard longbow and ultimately help Grug understand why, in his words, "long bow gooder than thick bow."

1.1 What exactly *are* we studying about bows?

After some quick internet scouring, we quickly discovered that bow efficiency is almost entirely a function of the bow body structure rather than the string used. Instead of tension on the string storing energy, we instead find that energy is stored in the act of deforming the bow, which is ultimately released back to its original shape when releasing the arrow.

1.2 Experimental Overview

Equipped with this knowledge, we begin our journey by defining in more concrete terms our problem: we aim to determine the relationship between a beam-shaped bow's dimensions and the strain energy it can store (and further, the displacement of an arrow fired by that bow). In order to do this, we vary two parameters: length, or vertical height, of the bow and the thickness of the bow.

Figure 1: Standards for variable naming conventions.

We will determine the effects of these variations on strain energy from the deformation of each bow. We will measure this by proxy, using the distance traveled by each arrow.

As a cool note, this problem can be as rigorous or as simple as we want it to be, depending on how we choose to model the beam deflection of the bow (i.e. selecting a non-linear function and using something out of the depths of Euler's *Elastica* or something simpler like the Euler-Bernoulli model we learned in class).

1.3 Why do we care about bows in the 21st century?

We chose this question because it touches on fundamental principles of energy transfer and mechanics, which are applicable not only in the context of traditional archery but also in various engineering fields. We are viewing this problem as an application of the culmination of the ME 80 coursework, applying beam deflection to a problem that involves motion and fun... The models we pose and test could have an impact in new fields of research regarding mechanotransduction (like what Ovi does), which makes significant use of AFM cantilever arms that experience similar forces and deformations to the bows we model (shoutout ME 244).

2 Design

As we mentioned above, this problem's difficulty is nearly infinitely scalable. We attempt to theoretically tackle two different versions: at a high level, one makes the same assumptions we've consistently made in class regarding beam deflections as linear models, and the other chooses to disregard those assumptions and more realistically model the bow's deformation as a nonlinear function.

2.1 Ideation and specific choices

For our CAD and prototype models, we designed a model of a bow and arrow that matches the level of theory, amount of physical material, and working equipment we're currently equipped with: BEAMS! Using a sheet of 6061-T6 tempered aluminum, we created rectangular strips to serve as the bows and developed two test suites of three bows each. The first suite varies in length (approximately in ratios of $2:3:4$) and the second varies in thickness (approximately in ratios of $1:2:4$). For drawstrings, we used braided fishing line rated for 300 pounds of pull. For arrows, we use 12-inch long wooden dowels that weigh approximately 5 grams each.

Traditionally, bows are strung while they are already under tension. To simplify our problem, we chose to string our bows while they were not under any load. This makes hand calculations and CAD/FEA simulations significantly easier and does *not* eliminate any of the fun of firing the bows (and with the math to come, we'd be grateful that we made this choice).

While many real-world bows are constructed from wood, carbon fiber, or other metals, our choice of aluminum allowed us to create a functional and testable prototype within our constraints without worrying about cost or non-isotropic behavior that is seen in other common bow materials such as wood (shoutout Jay who happened to have a sheet of Aluminum on hand).

2.2 CAD models

Our CAD models are one-to-one models of the real bows we intend to use. Because our bows are reasonable sizes, we see no reason to scale in CAD, as it is reasonable to manufacture full-scale bows. We will be analyzing the entirety of the bow in CAD – a deformed and loaded bow can essentially be simplified as two cantilevers joined by a fixed joint where the archer grips onto the handle.

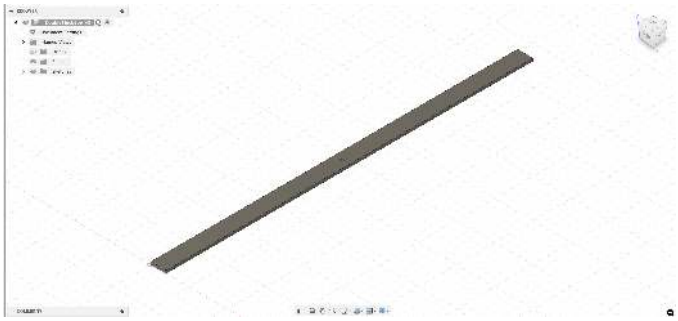


Figure 2: An example of the CAD bow with the hole in the middle for the arrow.

3 Hand calculations and basic simulations

The thorough derivations and interesting theory all lie in the appendix, and we please (pretty please) ask that you all take a look at the math we develop, as it's genuinely cool stuff (and it took a *lot* of hard work)!

3.1 The simple solution: linear beam theory

In the spirit of our course, we can somewhat generously model our bow as a linear deformation problem under Euler-Bernoulli conditions. While this may be untrue to the spirit of real life, this makes calculations *massively* easier (as we will soon see). Before we begin our calculations, we define the set of variables we will be working with in this section:

Symbol	Description	Values
w	width (in)	1.0625
h	height (in)	16.0625, 24.0625, 29.5
t	thickness (in)	0.09, 0.18, 0.36
θ	string-bow angle (deg)	30°, 45°, 60°
ϕ	launch angle (deg)	10°
E	elastic modulus (GPa)	68.9
I	inertia (in^4)	$\frac{wt^3}{12}$
m	mass of arrow (g)	5
T	tension on bow (N)	12.25, 24.5
H	height of archer (ft)	5

Assume the bow remains largely linear. Let the ends of the drawstring form angles θ with the bow. We pull back with force T on the string. Thus, at each string-bow interface, a total force $\frac{1}{2}T$ is applied in the direction of θ. We have then that $F_y = \frac{1}{2\sin\theta}T$. In solving for strain-energy, we consider the bow to be two cantilever arms joined at a fixed center point.

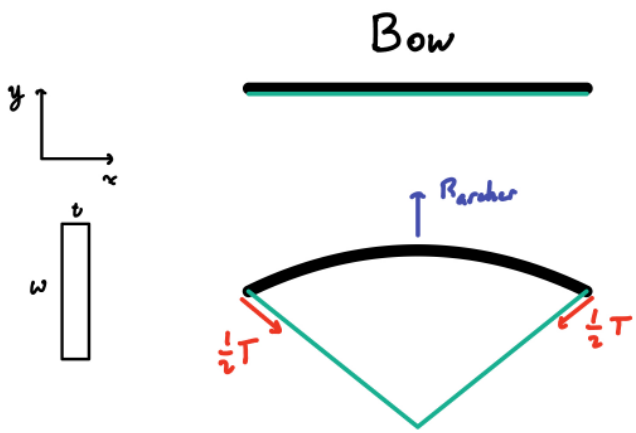


Figure 3: A free body diagram approximation of our bow problem.

While the FBD accentuates the bend, we will simply pretend it doesn't exist for this first set of calculations! Under linear beam conditions, we can use assumptions from our course and from the textbook to derive an expression for strain energy U_{total}. See Appendix Section 10.1 for a thorough derivation of the following equation:

$$\boxed{U_{\text{total}}(\theta, h, t) = \frac{T^2}{8Ew}\frac{h^3}{t^3\sin^2\theta}}$$

3.2 The complex solution: large deformation theory

If we want to model more true-to-life conditions, we drop the linear assumption and delve into the world of large deformations—a world we quickly grew to marvel at and despise at once... We detail thorough derivations and calculations in creating a second-order differential equation to represent a cantilever arm under a large deformation. See Appendix Section 10.2 for the full proof of this relationship, but here we simply state the result which is a relationship between the change of the angle θ, defined as the angle between the tangent vector to the bow and the horizon at a point s on the arc of the bow, and the force applied P:

$$\boxed{EI\theta''(s) + (s-L)P\cos\theta(s)\theta'(s) + P\sin\theta(s) = 0.}$$

Ultimately, we use the integral relationship between curvature κ and strain energy U_i detailed in Appendix Section 10.2 to see:

$$\boxed{U_{\text{total}} = 2U_i = 2P\int_0^L (\cos\theta(L) - \cos\theta(s))\,ds}$$

3.3 Kinematics

As detailed above, we assume that the arrow is a point mass in solving our dynamics equations. We also neglect the effects of aerodynamics and operate under vacuum conditions. Regarding energy transfer, we assume 100% of the strain energy in the bow is converted to kinetic energy in the arrow, neglecting any heat dissipation. Under these assumptions, we can begin the kinematic

analysis of the arrow. We will solve using a general U_{total} which can be obtained from either technique detailed above. Following the derivation detailed in Appendix Section 10.3, we find final x-displacement x_f as a function of U_{total} and ϕ:

$$\boxed{x_f(U_{\text{total}}, \phi) = \frac{U_{\text{total}}}{mg}\left(\sin 2\phi + 2\cos\phi\sqrt{\sin^2\phi + \frac{gHm}{U_{\text{total}}}}\right)}$$

3.4 Primary Analysis Criteria

As made clear through the derivations above, we intend to plot functions of U_{total} and x_f if possible, highlighting their cubic and inverse cubic relationships on increasing height and thicknesses, respectively. From FEA analysis, we ideally would be able to measure U_{total} directly (or at least beam deflection values, enabling us to integrate); however, as we'll soon see, Macbooks are apparently not cut out for Fusion. In the real-life test, there is no easy way to measure strain energy, so we will instead use x_f as our proxy measurement. Again, because the relationship between x_f and U_{total} is largely linear, we expect cubic and inverse cubic relationships to varying height and thickness, respectively.

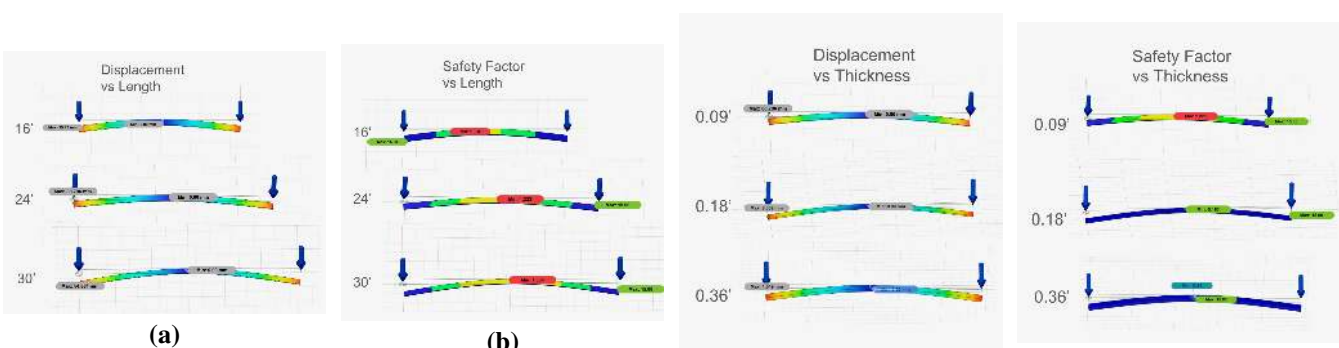
4 FEA Simulation Results

4.1 Initial assumptions and results

Six different bows were modeled as simple rectangular prisms. In three cases, the thickness was held constant (0.09') while the length was altered (16', 24', 30'). In the other three cases, the length was held constant (24') while the thickness was altered (0.09', 0.18', 0.36').

We used the static stress simulation workspace to model these bows. Since the drawstring in a standard firing configuration is pulled with a force from the very middle of the bow, we assumed that half of the pull force is applied at one end face of the bow and the other half is at the other end. There is a fixed constraint applied at the very center of the bow (where the archer would be holding it). Bows were modeled using 6061 aluminum, which is the same material we used to construct the real bows with.

This FEA does NOT take into account the fact that the elastic deformation of the bow is so large that the point in Cartesian space where the tensile force is applied changes location as the bow deforms. As far as we are aware, Fusion does not have the capability to model this changing boundary condition well.



(a) **(b)** **(c)** **(d)**

Figure 4: (a) Simulated displacement of beams w.r.t. changing length ($\max\delta \approx 20mm, 60mm, 55mm$*);
(b) Simulated Factor of Safety (F.S.) w.r.t. changing length ($\min F.S. \approx 1.87, 1.22, 1.03$*);
(c) Simulated displacement of beams w.r.t. changing thickness ($\max\delta \approx 60mm, 8mm, 1mm$);
(d) Simulated Factor of Safety (F.S.) w.r.t. changing length ($\min F.S. \approx 1.22, 5.16, 12.8$).
**To prevent yielding, the longest bow undergoes exactly half the force of the other bows; in determining graphical relationships, this will be accounted for.*

4.2 Sanity check

Nonetheless, after running the simulation, the maximum displacement from the endpoints of the bow seem to match the measured deformation from running tests in real life. The thickest bow (0.36") deforms only 1mm at its ends, while the thinnest bow (0.09") deforms around 60mm. Furthermore, since there is a factor of 4 difference between their thickness, our mathematical model expects a cubic change (so 64x) in maximum deformation, which our FEA model roughly confirms. See the complete results below.

Parameter	0.09' Thick Bow	0.18' Thick Bow	0.36' Thick Bow
Min Safety Factor (at center)	1.22	5.16	13.66
Does it Yield?	No	No	No
Max Deformation (at ends), mm	1.05	8.31	60.80
Real Deformation Ratio (compared to 0.09' bow)	1	7.91	57.9
Expected Deformation Ratio (compared to 0.09' bow)	1	8	64
Error (%)	N/A	-1.13	-9.53

Table 1: Qualitative and quantitative analysis of FEA simulation on varying thickness.

Parameter	16' Long Bow	24' Long Bow	30' Long Bow*
Min Safety Factor (at center)	1.87	1.22	1.03
Does it Yield?	No	No	Very Close
Max Deformation (at ends), mm	19.43	60.80	54.66
Extrapolated	—	—	109.32
Real Deformation Ratio (compared to 16' bow)	1	3.13	5.66
Expected Deformation Ratio (compared to 16' bow)	1	3.375	6.59
Error (%)	N/A	-7.26	-14.11

Table 2: Qualitative and quantitative analysis of FEA simulation on varying length.

4.3 Limitations

Another note is that we originally intended to use Fusion FEA to analyze the strain energy stored from the deformation in each load case. Unfortunately, we realized that Fusion does not have a feature to calculate strain energy.

Afterward, we aimed to use the FEA stress/strain data to calculate the strain energy using the following equation:

$$\boxed{U_i = \frac{1}{2}\int_V \sigma\varepsilon dV} \tag{1}$$

where V is the volume of the bow, σ is stress, and ε is strain.

However, even though Fusion lets you apply surface probes to find stress and strain at individual points, there is no way to export a data file that shows the stress/strain across the entire mesh. Therefore, using Fusion to find the strain energy due to deformation is impossible. More advanced analysis software such as ANSYS would be required for proper analysis of strain energy.

5 Prototype Testing

5.1 Manufacturing

In order to conduct experiments on bows, we need—you guessed it—some physical bows! Using the sheet metal shear, we cut many individual 24-in long and 1 in-wide bars from a sheet of 24in by 48in by 0.09in 6061-T6 aluminum. In order to vary length, we cut one bar into a 16in-long bow. To make the "long" bow, we made a 1.0-in wide cut in the other direction of the sheet (which we admittedly should have done *before* we cut all of the other bars).

In order to emulate varying thickness without welding pieces together or buying a thicker block of aluminum, we pop-rivetted stacks of two and four bars.

To all of the bows, we made horizontally-centered small holes exactly 0.25in from each end to serve as holes for the drawstring, which was tied such that the string was entirely tense and flush with the vertical, non-deformed bow.

For the arrows, we used 12" length, 1/4" diameter wooden dowels with a slit cut into one end to attach to the string. These dowels weighed roughly 5 grams and were retrieved from AMPS.

See Figures 8 and 9 below for real-life visuals of the bows!

5.2 Experiment Set-up, Conditions, and Limitations

Before outdoor firing-range-style testing began, we had to make sure we'd be firing the bows with a fixed pull-back force. At first, we tested using the force gauge to pull a steady 24.5N (displayed as 2.5 kg), then performing a sort of "hand-off" from the hooked force gauge to the slitted arrow. This proved to be very impossible to do consistently. However, we noticed something

extremely useful: for a given bow, we were pulling back the arrow almost exactly the same amount for the required force. Thus, we realized it would be much easier to simply mark on the wooden dowel where it should be pulled back to. We performed this pull-back-and-mark technique 5 times per bow to ensure our hunch was correct—and it was! All five marks would be within approximately 1mm of each other. This eliminated a *massive* source of user error almost instantaneously.

Figure 5: Marking the dowels at the pullback distance that corresponds to2.5kg (24.5N) force.

On an unfortunately windy Tuesday, we decided it was time to launch our bows. At the Columbae front lawn, we set out a 25ft tape measure exactly 25ft from where we planned to fire the bows. During our extremely rigorous testing, we fired 12 shots per bow. Each shot was fired from approximately 1m off the ground at an angle of 10 degrees above horizontal. We measured the horizontal range of the arrow after each shot, and the results are plotted below. The experimental results correlate well with our calculated and simulated theoretical results, following a largely cubic/inverse cubic trend.

Figure 6: Leg-ethan-olas Himself. In the flesh. # ArcheryIsBack.

Regarding kinematics, real-world testing involved uncontrolled factors such as energy loss due to heat, noise, and vibration of the bow. Furthermore, the day of testing was partly windy. Thus, we expected to see noticeably lower x-displacement values across the board than were predicted by our Python model.

5.3 Results and Comparisons

We found that our experimental results correlated well with our calculated and simulated theoretical results. There is a roughly cubic relationship between the length of the bow and the horizontal range and a roughly inverse cubic relationship between the thickness of the bow and the horizontal range.

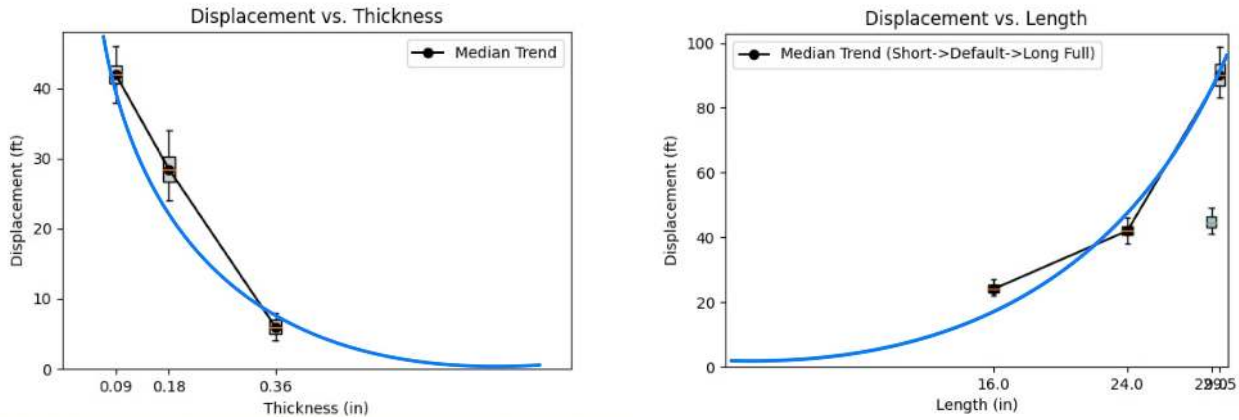


Figure 7: Left: Real displacement vs thickness fitted to an inverse cubic function. Right: Real displacement vs length fitted to a cubic function*.
**We adjust the light blue values of the longest bow, which were measured at half the tension applied to the other bows by doubling displacement (which can be justified by following our governing equations)*

There were quite a few limitations in our testing process. Since we only had access to 0.09" thick sheet metal, we had to approximate different thicknesses by stacking and riveting multiple layers together, which may not have perfectly replicated the properties of a single, uniformly thick piece. Additionally, our bows had varying stiffnesses, which made it difficult to apply the same pull-back force across all tests. To address this, we adjusted the force gauge settings based on each bow's behavior, ensuring that every test produced meaningful results while still maintaining a standardized approach. For the 0.36" thick bow, we increased the loading force to 5 kg instead of 2.5 kg because it wouldn't shoot properly otherwise. Conversely, for the longer bow, we reduced the force to 1.25 kg, as our arrows weren't long enough to accommodate the extended draw length created by a 2.5 kg force.

Another limitation was the accuracy of our distance measurements. Conducting tests on a grass field with a tape measure introduced potential inconsistencies due to factors such as wind, uneven terrain, and irregularities in arrow flight caused by drag and instability. Variability in landings also made precise measurements challenging. To mitigate these issues, we conducted twelve test shots per arrow and averaged the results, helping to minimize the effects of external factors.

6 Discussion

6.1 What We Learned

Intuitively (at the so-called Grug level), it would seem that longer and skinnier bows will deform more than shorter, thicker bows. We validated this with the mathematical model, FEA simulation, and real-world testing to show that there is in fact a cubic relationship between the bow's length and its strain energy via bending and an inverse cubic relationship between the bow's thickness and its strain energy via bending. We were able to validate this relationship experimentally through real-world testing (Figure 7). Furthermore, FEA simulations demonstrated a near cubic relationship between the thickness of the bows and the amount of maximum deflection at the ends. All three methods (hand calculations, FEA, and experimental testing) indicate consistent matching results for how the length and thickness of a bow affect its performance.

6.2 Comparing FEA Simulations to Real Results

There were notable differences in the expected deflection from hand calculations, and the deflections predicted by FEA. In particular, for the longest bow case (30" length), the deflection predicted by FEA (55mm) was far less than the real measured deflection (140mm). We attribute this to the inaccuracies of our FEA modeling – the location at which the forces were applied in the simulation stayed constant despite the fact that the physical ends of the beam already moved drastically. Fusion does not have a suitable method for modeling cases with large deformations. It generally assumes linear deformation (Euler-Bernoulli style). This could be improved by using different analysis software that correctly models large deformation cases. However, Fusion was able to correctly analyze cases with smaller deformations – these had much more accurate results that were comparable to our real-world testing.

6.3 Planned v.s. Reality

One surprising factor that was not taken into account while modeling the bows *or* building them was the fact that in our longest bow case (30" length), applying the same force of 24.5N to the bowstring would result in such large beam deflection that the arrow was too short to contact the bowstring and the bow at the same time. Thus, we had to halve the force to 12.5N for this case only.

Furthermore, another oversight that we missed while planning was that certain sizes of bows do not lend themselves well to launching arrows due to physical constraints. We initially created a really short bow (8” long), but it deformed so little that it could not even launch an arrow even under very high loads. Furthermore, some of the thinner bows experienced yielding when too much force was applied on the string; hence the reason for the 24.5N max limit (this amount prevents yielding for all bow sizes that we ended up actually using).

The original bows were designed to have arrows being launched while being held against the side of the bow. This resulted in arrows that would often launch more sideways than forwards. We corrected this by drilling a hole in the center of each bow for the arrow to slot through and launch out of. This ensured that each arrow would fire directly forwards out of the bow. However, this introduces a new potential limitation: increased friction at the bow-arrow interface. However, we think this was generally negligible in terms of observing trends.

Finally, in the planning stage, we assumed that the math for deriving a relationship between the bow geometry and its associated strain energy would be relatively simple. However, in order for the math to be simple (using equations from Chapter 14 of the textbook), you must assume very small deformations ($<< 10\%$). As described elsewhere in this paper, beam deflection in bows is very large, so these assumptions break down – much more complicated mathematical calculations had to be conducted to better model the mechanics of what is truly occurring.

7 Final Products

Our final product consists of six aluminum bows with varying lengths and thicknesses, along with arrows for demonstration and testing purposes:

- Three bows that vary in length (16”, 24”, 30”).
- Three bows that vary in thickness (0.09”, 0.18”, 0.36”).
- Bowstrings of braided no-stretch fishing line rated for 300lb

Figure 8: The bows of different thicknesses with standard length of 24” and width of 1”. Thicknesses: 0.36”, 0.18”, and 0.09” as shown from left to right.

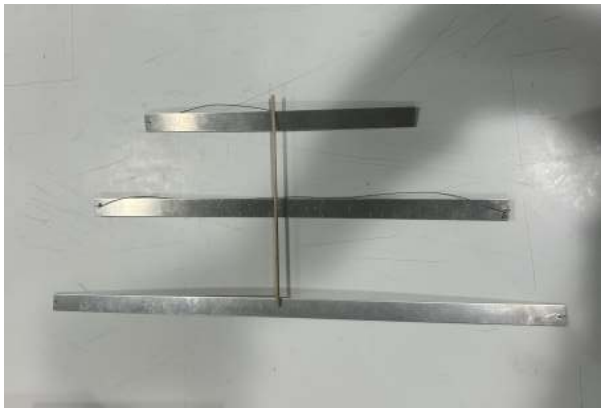


Figure 9: The bows of different lengths with standard thickness of 0.09” and width of 1”. Lengths: 16”, 24”, and 30” as shown from top to bottom.

8 Conclusion

8.1 What should we have changed

The process conducted on this project was rigorous, but not without flaws. If this project were to be re-done, there are several things that would have been done differently.

- Bows would have been designed with a higher safety factor to prevent accidental yielding during testing. It also might have been better to construct them from steel instead of aluminum.
- A more advanced FEA software would have been used as opposed to Fusion (see below).
- Thicker bows would have been manufactured as solid prisms rather than a stack of thin sheets riveted together.
- Testing of launching the bows would have been done indoors to prevent the effects of wind on the arrows (which was quite significant).
- Several more sizes of bows would have been made to have more data points to determine the experimental (not theoretical) relationship between bow thickness, bow length, and arrow range.

8.2 What should we keep the same

- The hand-calculated mathematical analysis resulted in a relationship between strain energy, bow thickness, and bow length that was rigorous, reasonable, and consistent.
- The CAD of the bow design was quite simple and easy to manufacture.
- The Python and MATLAB models were robust and created plots for predicted bow strain energy and arrow range that line up with expected results.

8.3 Concepts we have internalized

Throughout this project, intuition about beam deformation theory was gained by all members of the group. We were able to internalize the mathematical relationship between beam deflection, beam thickness, and beam length in both linear and non-linear loading conditions. An actual mathematical relationship was derived to validate our intuition that the longer and thinner a beam is, the more it will deform.

8.4 If we had two more weeks...

There are several potential paths to expand on the results from this project if we had more time. First of all, redoing the FEA in a different analysis software such as COMSOL or ANSYS would have allowed for better modeling of the bow under non-linear loading conditions (the position at which the forces are applied changes as the beam is deformed).

Furthermore, another interesting pathway to further develop this project is to develop a mathematical optimization problem that takes three different factors into consideration:

1. Maximizing the strain energy via the deformation of the bow.
2. Keeping the applied force under a reasonable limit that a human could apply to the bowstring (perhaps 100N).
3. Preventing yielding (F.S. $\geq$ 1.5).

Combining these three metrics into a optimization problem could result in a theoretical “perfect” longbow design by giving us three exact numbers for the bow length, thickness, and width that maximizes the range of the arrow.

9 Peer Reviews

NOTE: Because the mathematical analysis portion of this project ended up being far heavier than expected, the initial designation of roles was not exactly followed. Every member played a significant role in every part of the project (the closest approximation that we can "give" would be lumping CAD and FEA into one role and giving the math analysis and MatLab/Python simulation to the original FEA person).

9.1 Peer Review for CAD

The CAD of the bows is relatively straightforward – it is merely an extruded rectangular prism. This simplifies the manufacturing process to being able to aluminum sheet metal on an electric shear. It is also quite simple to conduct FEA on these models. However, one minor discrepancy is that forces need to be applied to specific model features (faces, edges, etc) in the FEA simulation workspace. In the CAD model, the drilled holes where the bowstring was tied through were not present. Thus, the forces were approximated to be applied to the edge faces of the bow rectangular prism instead of at the exact location of the holes. This could have potentially resulted in some minor discrepancies – the locations of the FEA forces were about ½” past their true location on the real bows. Overall, the CAD models were easy to manufacture and conduct FEA on.

9.2 Peer Review for FEA

The FEA was done assuming a linear beam with small beam deflection. However, this assumption is not truly accurate, as experimental testing with the real bows show that the predicted results from FEA are not truly accurate. The maximum experimental deflection for the long and thin bow case is greater than the predicted deflection in Fusion.

It also would have been convenient to access a .txt or .csv file with the FEA results for stress and strain so that deformation energy could have been calculated by integrating the product of stress and strain across the volume of the bow. This way, it would have been easier to compare the predicted energy stored by the bow against real world testing (where the experimental range of the arrow is a proxy for the energy stored). This perhaps could have been completed with more robust FEA software. However, I acknowledge that is out of the scope of this course.

9.3 Peer Review for Manufacturing

The accuracy of the real world experimental data might have been slightly affected by the fact that the "thick" bows were just multiple "thin" bows riveted together. This is not exactly a direct analog to the solid thick bow that was modeled in CAD – there may be possible boundary effects (sliding, non-uniform bending among different sheets, etc) between each thin sheet that we did not account for in CAD. We only had 0.09' thick aluminum sheet available, so the only way to construct thicker bows like those in CAD is to stack multiple together. In a perfect world, the thicker bows would have been manufactured with thicker sheet aluminum to mitigate potential effects from the boundaries between the multiple sheets and/or the rivets.

Furthermore, when the bowstring was tied to the ends of each bow, there was some slack left in each string at the bow's neutral position. Thus, even under the same central string pull force, the angle of the string tensile vectors still were not exactly consistent between the different bows, because some bowstrings had more slack than others. It would have been better if each bow was strung at the same tension before being loaded.

Finally, when launching the arrows, they tended to not fly straight – the arrows flip and spin in the air, creating unpredictable aerodynamic conditions that could have dramatically affected the results. This could have been resolved by adding feathers to each arrow. Nonetheless, the resulting bows were robust and fun to shoot. They adequately launched arrows to give meaningful real-world data.

10 Appendix

10.1 Computing Strain Energy — Euler-Bernoulli Beam Theory

10.1.1 Calculating moment

To begin, we assume our bow to be two cantilevers joined at a fixed center. We aim to derive a function $M(x)$ that describes the internal moment of the beam at a point x (with $x = 0$ being at the free end of the beam, where the force is being applied, and $x = L = \frac{h}{2}$ being the center of the bow). By inspecting a section of the bow of length x and assuming we are at static equilibrium, we observe that

$$\begin{aligned} \sum_i M_i &= 0 \\ M(x) + xF_y &= 0 \\ M(x) &= -xF_y. \end{aligned}$$

From our textbook, we define strain energy U_i as follows:

$$U_i = \int_0^L \frac{M^2(x)}{2EI} dx.$$

Substituting in the values that we have, we see:

$$\begin{aligned} U_i &= \int_0^L \frac{(xF_y)^2}{2EI} dx \\ &= \frac{F_y^2}{2EI} \int_0^L x^2 dx \\ &= \frac{F_y^2 L^3}{6EI} \end{aligned}$$

Thus, we have an expression for total strain:

$$U_{\text{total}} = 2U_i = \frac{F_y^2 L^3}{3EI}.$$

Plugging in expressions for I and substituting $h = 2L$ and $F_y = \frac{T}{2\sin\theta}$, we see:

$$\frac{F_y^2 L^3}{3EI} = \frac{T^2(\frac{L}{2})^3}{12E\frac{wt^3}{12}\sin^2\theta}$$

$$\boxed{U_{\text{total}}(\theta, h, t) = \frac{T^2}{8Ew}\frac{h^3}{t^3\sin^2\theta}} \tag{2}$$

10.2 Computing Strain Energy — Non-linear Deformation

In the name of curiosity, we've taken it upon ourselves to develop an applicable form of non-linear beam theory to fit our bow scenario. The gains in a more accurate model for an experiment as simple as ours may not be worth it time-wise, but we were curious enough to take the bait and dive into Euler's *Elastica* (and its corresponding course notes online from MIT and UMich).

10.2.1 Developing curvature

Before exploring a more complex mathematical model of our bows, it is worth introducing the concept of curvature κ. We first set out to relate the angle of the tangent vector to the curve to deflection of the bow. We begin this by considering an arbitrary curve $y = w(x)$ where w represents the deflection function across the plane. We compute the unit tangent vetor at a point x:

$$\begin{aligned}
\vec{t}(x) &= [1, w'(x)] \\
\vec{\hat{t}}(x) &= \frac{1}{\|\vec{t}(x)\|}[1, w'(x)] \\
&= \frac{1}{\sqrt{1 + (w'(x))^2}}[1, w'(x)].
\end{aligned}$$

Now, we can compute the tangent angle $\theta(x)$, defined as the angle between the unit tangent vector and the horizontal:

$$\begin{aligned}
\tan\theta(x) &= \frac{\frac{w'(x)}{\sqrt{1+(w'(x))^2}}}{\frac{1}{\sqrt{1+(w'(x))^2}}} \\
&= w'(x) \\
\theta(x) &= \arctan w'(x).
\end{aligned}$$

Now, we introduce the concept of curvature κ, which we define as the change in θ with respect to arc length s, which is notably different than regular displacement x:

$$\kappa = \frac{d\theta}{ds}.$$

. We can derive an expression for the differential arc length $ds := \sqrt{dx^2 + dy^2}$ in terms of w as follows:

$$\begin{aligned}
y &= w(x) \\
dy &= \frac{dw}{dx} = w'(x)dx \\
ds &= \sqrt{dx^2 + dy^2} \\
&= \sqrt{dx^2 + (w'(x)dx)^2} \\
&= \sqrt{1 + (w'(x))^2}dx.
\end{aligned}$$

Now, we use the chain rule on the expression for curvature:

$$\begin{aligned}
\kappa &= \frac{d\theta}{ds} \\
&= \frac{d\theta}{dx}\frac{dx}{ds} \\
&= \frac{d\theta}{dx}(\frac{ds}{dx})^{-1} \\
&= \frac{d\theta}{dx}\frac{1}{\sqrt{1 + (w'(x))^2}}.
\end{aligned}$$

Lastly, we differentiate $\theta(x) = \arctan w'(x)$ with respect to x:

$$\begin{aligned}\kappa &= \frac{d\theta}{dx}\frac{1}{\sqrt{1+(w'(x))^2}} \\ &= \frac{w''(x)}{1+(w'(x))^2}\frac{1}{\sqrt{1+(w'(x))^2}}\end{aligned}$$

$$\kappa = w''(x)\left(1+(w'(x))^2\right)^{-\frac{3}{2}}. \tag{3}$$

10.2.2 Deriving the nonlinear beam equation

As we have defined the exact expression for curvature, we can now move on to deriving the nonlinear beam equation that Euler so graciously created for us 300 years ago. However, here, we're doing it Eu-ethan style (pronounced "Oithan"). We begin with the relationship between bending moment and curvature in linear elastic materials (courtesy of an MIT course note):

$$M(s) = EI\kappa(s).$$

Where E is Young's modulus and I is the cross-sectional moment of inertia. Now, using the original definition of curvature, we substitute into our equation above:

$$M(s) = EI\frac{d\theta}{ds}.$$

Now, we can begin with classic statics by assuming static equilibrium. As we do in the linear case, we will model the bow as two fixed cantilevers joined at the center. From now on, we will only be working with half the bow (arc length $L := \frac{h}{2}$ with a single load of $P := \frac{T}{2}$ at the edge $s = 0$) until computing total strain energy. We will first derive an expression for the moment of the cantilever arm at a point on the arc s after applying a vertical load. To do this, we will assume that the force at the cantilever end is perpendicular to the arm. While this assumption may not be true to real life, I fear that I'm already too deep in treacherous waters, and I'd like to make my life very slightly easier:

$$\begin{aligned}M(s) &= -\sum_i \text{distance} \times \text{force} \\ &= (L-s)\sin\theta(s)\cdot P.\end{aligned}$$

We can now substitute back into our expression above to see:

$$(L-s)\sin\theta(s)\cdot P = EI\frac{d\theta}{ds}.$$

This equation tells us about the relationship between moment and curvature at a point. However, we seek to understand how this relationship *changes* as we move across the beam. Thus, we differentiate this equation with respect to arc length s:

$$\begin{aligned}\frac{d}{ds}\big((L-s)\sin\theta(s)\cdot P\big) &= \frac{d}{ds}\big(EI\frac{d\theta}{ds}\big) \\ EI\frac{d^2\theta}{ds^2} &= P\left(-\sin\theta(s) + (L-s)\cos\theta(s)\frac{d\theta}{ds}\right).\end{aligned} \tag{4}$$

Now, consider an element of the beam ds. At this point, there are several forces acting at once: an internal normal force $N(s)$, an internal shear force $V(s)$, and the external load P. Because we are at static equilibrium, in the direction perpendicular to the beam, we can write the following expression:

$$(V(s) - V(s+ds)\cos d\theta) + N(s+ds)\sin d\theta = 0.$$

We can abuse the small angle approximation here to argue that $\cos d\theta \approx 1$ and $\sin d\theta \approx d\theta$ to simplify our equation:

$$\begin{aligned}(V(s) - V(s+ds)) + N(s+ds)d\theta &= 0 \\ V(s+ds) - V(s) &= N(s+ds)d\theta \\ \frac{V(s+ds)-V(s)}{ds} &= N(s+ds)\frac{d\theta}{ds}.\end{aligned}$$

Now, we let $ds \to 0$. By the limit definition of a derivative, we see:

$$\frac{dV}{ds} = N(s)\frac{d\theta}{ds}. \tag{5}$$

With equation 3, we can now relate this to the external load. According to simple force balance, we should have that $V(s) = P\cos\theta(s)$ for $V(s)$ perpendicular to the cantilever. We can substitute this in:

$$\begin{aligned}\frac{d}{ds}\left(P\cos\theta(s)\right) &= N(s)\frac{d\theta}{ds} \\ -P\sin\theta(s)\frac{d\theta}{ds} &= N(s)\frac{d\theta}{ds}.\end{aligned}$$

Assuming that $\frac{d\theta}{ds}$ is nonzero, we see that

$$N(s) = -P\sin\theta(s). \tag{6}$$

We also consider static moment equilibrium about the center of curvature, which we assume to be a radius R away from all points on the bow:

$$\begin{aligned} R \times V(s) &= (Rd\theta) \times N(s) \\ V(s) &= N(s)\frac{d\theta}{ds}. \end{aligned}$$

Substituting back into the expression for shear force perpendicular to the bow at point s, we now have

$$\begin{aligned} V(s) &= P\cos\theta(s) \\ &= N(s)\frac{d\theta}{ds}, \end{aligned}$$

giving us

$$\cos\theta(s) = \frac{N(s)}{P}\frac{d\theta}{ds}. \tag{7}$$

Now we have all of the tools ready to finish our derivation. We begin by plugging what we have into equation 2:

$$\begin{aligned} EI\frac{d^2\theta}{ds^2} &= P\left(-\sin\theta(s) + (L-s)\cos\theta(s)\frac{d\theta}{ds}\right) \\ EI\frac{d^2\theta}{ds^2} &= -P\sin\theta(s) + (L-s)P\cos\theta(s)\frac{d\theta}{ds}. \end{aligned}$$

By invoking equation 5, we see

$$EI\frac{d^2\theta}{ds^2} = P\left(-\sin\theta(s) + (L-s)\cos\theta(s)\frac{d\theta}{ds}\right)$$

Therefore, we are left with the following second order differential equation:

$$\boxed{EI\theta''(s) + (s-L)P\cos\theta(s)\theta'(s) + P\sin\theta(s) = 0.} \tag{8}$$

10.2.3 Deriving an equation for strain energy

The strain energy U_i is given by

$$U_i = \frac{1}{2}\int_0^L M(s)\cdot\kappa(s)ds.$$

We can begin substituting in expressions from our previous derivations:

$$\begin{aligned} U_i &= \frac{1}{2}\int_0^L EI\kappa(s)\cdot\kappa(s)ds \\ &= \frac{1}{2}\int_0^L EI\left(\frac{d\theta}{ds}\right)^2 ds. \end{aligned}$$

Using our approximation of a nonlinear beam, we can begin by expanding on equation 6:

$$\begin{aligned} EI\frac{d^2\theta}{ds^2} &= -P\sin\theta(s) \\ EI\frac{d^2\theta}{ds^2}\frac{d\theta}{ds} &= -P\sin\theta(s)\frac{d\theta}{ds} \\ \frac{EI}{2}\frac{d}{ds}\left(\left(\frac{d\theta}{ds}\right)^2\right) &= -P\frac{d}{ds}\left(\cos\theta(s)\right). \end{aligned}$$

Now, we can integrate both sides with respect to s:

$$\frac{EI}{2}\left(\frac{d\theta}{ds}\right)^2 = -P\cos\theta(s) + C \tag{9}$$

for some arbitrary constant C.

Now, we must define the rest of our boundary conditions. We already described that at the free end $s = L$, we have $\frac{d\theta}{ds}\big|_{s=L} = 0$. Thus, we see

$$
\begin{aligned}
0 &= -P\cos\theta(L) + C \\
C &= P\cos\theta(L).
\end{aligned}
$$

Substituting this in, we observe:

$$
\begin{aligned}
\frac{EI}{2}\left(\frac{d\theta}{ds}\right)^2 &= -P\cos\theta(s) + P\cos\theta(L) \\
&= P\left(\cos\theta(L) - \cos\theta(s)\right).
\end{aligned}
$$

Now, to solve for the differential term:

$$
\frac{d\theta}{ds} = \left(\frac{2P}{EI}\cos\theta(L) - \cos\theta(s)\right)^{\frac{1}{2}} \tag{10}
$$

Ultimately, we can the following expression for strain energy:

$$
\begin{aligned}
U_i &= \frac{1}{2}\int_0^L EI\left(\frac{d\theta}{ds}\right)^2 ds \\
&= \frac{1}{2}\int_0^L EI \cdot \frac{2P}{EI}\left(\cos\theta(L) - \cos\theta(s)\right) ds \\
&= \int_0^L P\left(\cos\theta(L) - \cos\theta(s)\right) ds
\end{aligned}
$$

Therefore, we conclude with an expression for the total strain energy in the bow:

$$
\boxed{U_{\text{total}} = 2U_i = 2P\int_0^L \left(\cos\theta(L) - \cos\theta(s)\right) ds} \tag{11}
$$

To solve this integral fully, we must numerically calculate a solution to equation 6, as no closed-form solution has been discovered yet.

10.3 Computing final displacement of the "arrow"

First, with the assumptions detailed in Section 3.3 we set up an energy balance equation to solve for the initial velocity of the arrow:

$$
\begin{aligned}
\Delta U + \Delta KE &= 0 \\
U_{\text{total}} &= \frac{1}{2}mv_0^2 \\
v_0 &= \sqrt{\frac{2U_{\text{total}}}{m}}.
\end{aligned}
$$

Using the launch angle ϕ measured from the horizon, we can determine the x and y velocity components:

$$
\begin{aligned}
v_{0,x} &= v_0\cos\phi \\
v_{0,y} &= v_0\sin\phi.
\end{aligned}
$$

Assuming constant gravity $g := 9.82\frac{m}{s^2}$, we can determine the exact time that the arrow stays in the air:

$$
\begin{aligned}
y(t) &:= -\frac{1}{2}gt^2 + v_{0,y}t + H \\
y(t_f) &= 0 \\
y(t_f) &= -\frac{1}{2}gt_f^2 + v_{0,y}t_f + H = 0.
\end{aligned}
$$

By the quadratic formula:

$$
t_f = \frac{v_{0,y} \pm \sqrt{v_{0,y}^2 + 2gH}}{g}
$$

We trivially take the positive solution $t_f = \frac{v_{0,y}+\sqrt{v_{0,y}^2+2gH}}{g}$. Now, we can compute horizontal displacement by plugging t_f into the x-displacement function to see:

$$\begin{aligned} x &= v_{0,x} \cdot t_f \\ &= v_0 \cos\phi \cdot \frac{v_0 \sin\phi + \sqrt{(v_0 \sin\phi)^2 + 2gH}}{g} \\ &= \frac{v_0 \cos\phi}{g} \left(v_0 \sin\phi + \sqrt{v_0^2 \sin^2\phi + 2gH} \right). \end{aligned}$$

Substituting our expression for v_0, we have

$$\begin{aligned} x_f &= \frac{\sqrt{\frac{2U_{\text{total}}}{m}} \cos\phi}{g} \left(\sqrt{\frac{2U_{\text{total}}}{m}} \sin\phi + \sqrt{\frac{2U_{\text{total}}}{m} \sin^2\phi + 2gH} \right) \\ &= \frac{\sqrt{\frac{2U_{\text{total}}}{m}} \cos\phi}{g} \left(\sqrt{\frac{2U_{\text{total}}}{m}} \sin\phi + \sqrt{\frac{2U_{\text{total}}}{m}} \sqrt{\sin^2\phi + \frac{2gHm}{2U_{\text{total}}}} \right) \\ &= \frac{2U_{\text{total}} \cos\phi}{mg} \left(\sin\phi + \sqrt{\sin^2\phi + \frac{gHm}{U_{\text{total}}}} \right) \end{aligned}$$

Therefore, we have final x-displacement as a function of U_{total} and ϕ:

$$\boxed{x_f(U_{\text{total}}, \phi) = \frac{U_{\text{total}}}{mg} \left(\sin 2\phi + 2\cos\phi \sqrt{\sin^2\phi + \frac{gHm}{U_{\text{total}}}} \right)} \tag{12}$$

10.4 Implementing the Nonlinear Beam Equations in `Matlab`

Recalling from our derivation that the governing equation for a single cantilever can be written in the form

$$EI\,\theta''(s) + (s-L)\,P\,\cos\theta(s)\,\theta'(s) + P\,\sin\theta(s) = 0, \tag{13}$$

the MatLab code added at the end of the PDF sets up a finite element framework to (unfortunately incorrectly) numerically solve this same relationship. I (Ethan) ran out of time on this part, so I will give y'all my incorrect implementation in hopes that the spirit and vibes are good enough to get my point across. I first tried some straight-up solving of the massive ODE using ode45 but that unfortunately yielded bogus results. I moved onto doing some janky StackOverflow-based compilation of finite element analysis. Didn't really work but the theory's cool! Back to formal language... The implementation works as follows:

1. Geometry and Material Setup: The script begins by specifying the bow dimensions (w, h, t) along with the elastic modulus E and load $P := \frac{T}{2}$. This matches the same physical parameters assumed in our derivation, where h is the total height of the bow and T is the total tension.
2. Discretization via Hermite Elements: To capture the beam curvature $\theta(s)$ and its derivative, my script uses a cubic Hermite shape function. Unlike linear or quadratic interpolants that may only match function values at the nodes, cubic Hermite functions are designed to interpolate both the function and its first derivative. Each finite element has four degrees of freedom—θ and θ' at both nodes—which accommodates for the bending kinematics from our derivation.
3. Newton-Raphson Iteration: We note that the governing equation is nonlinear due to the appearance of $\sin\theta$ and $\cos\theta$ terms. Thus, the script assembles and solves a global system of equations via the Newton-Raphson iterations (which I have learned about at roughly 2:10 am on Tuesday March 18). This iterative approach refines the solution for $\theta(s)$ until the residual value is below a specified tolerance.
4. Strain Energy and Arrow Kinematics: Once $\theta(s)$ is obtained, the code computes the strain energy

   $$U_{\text{total}} = 2\int_0^L (\cos\theta(L) - \cos\theta(s))\,P\,ds,$$

   using our pre-derived functions. We can then follow energy balance and basic kinematics as detailed above. I don't use the discrete formula in my script to avoid parsing for typos/massive bugfixing (again, I am tired haha).

10.4.1 Parameterized Model for Comparison

In addition to the finite element solution, I have a more simple parametrized model for $\theta(s)$:

$$\theta(s) = \theta_{\max} \left(1 - \left(\frac{s}{L} \right)^2 \right),$$

where $\theta_{\max}$ is determined by matching the computed strain energy to an expected value of roughly 0.8 Joules (which we get by doing a quick back-of-the-napkin calculation from our real life values). This approach provides a simpler, but less accurate approximate, estimate of $\theta(s)$ and serves as a benchmark for the (wildly incorrect) finite element model.

I am sad to say I could not actually solve my ODE. It pains my heart, but I tried my darndest <3. I'm sorry Euler.